

Data recovery (also called data carving) tools aim at recovering the data contained in a forensic image, which may have been lost due to one or more of the following reasons:

- Deletion/corruption of the file/directory containing the data.
- Corruption of the underlying filesystem.
- Deletion/repartitioning of the partition containing the data.
- File containing the data is embedded into another data file.
- The extension has been changed for the file containing the data.

Since the associated meta-data structures are overwritten in all these scenarios, data cannot be identified or extracted in the regular way, and hence is called lost. To recover such lost data, usually certain content/attributes related to the data (or files containing the data) are given as an input to data recovery tools. The tools then scan the provided forensic image to try and find the required data. Data recovery tools play an important role in most forensic investigations, because smart malicious users will always try to delete evidence of their unlawful acts. Some important data recovery tools provided by BackTrack are described below. To know more, refer to <http://digital-assembly.com/technology/research/pubs/ieee-spm-2009.pdf>.

Foremost

Foremost is one of the popular data recovery tools. It can recover deleted data files of a particular type, from a forensic image acquired via tools such as dd. The foremost method of recovering deleted file(s) is based on the internal structure of a file, which usually includes attributes such as unique signatures, file headers, file footers, etc. For example, given a header and footer for a particular type of file to be recovered, it will start reading memory blocks from a media image. Once a matching header is found, the header, and the data following the header, is written into a separate file, until the matching footer is found, or a given file-size limit is reached.

However, due to its extraction procedure, Foremost has limitations when the files are fragmented, when the file header is over-written, when it is a common string, or is changed due to operations such as compression.

Foremost has a number of built-in extraction routines, which are meant for recovering a variety of file types, such as JPG, PNG, BMP, etc. However, if one cannot find built-in support for extracting a particular type, the Foremost configuration file is handy. You can easily tweak *foremost.conf* to include support for a specific file type.

Usage

Usage is as *foremost <options>*, where some of the important options are:

- s <number>**: This skips <number> blocks in the input image before beginning search of headers.
- i <image>**: This command specifies the input image or partition for Foremost.
- o <out>**: It specifies the output directory for storing recovered files. In case you are searching a live partition, do not specify an output directory that is on the input partition.
- c <config_file>**: Specifies the configuration file to be used.
- b <block-size>**: This specifies the block size to be used; the default is 512.
- t <type1, type2>**: This specifies (comma-separated) the different file types to be extracted.
- w**: This is to write an audit file, *audit.txt*, containing relevant recovery information, in the output directory—but do not write recovered files into the output directory.
- q**: It enables quick mode; searches are performed on block boundaries.

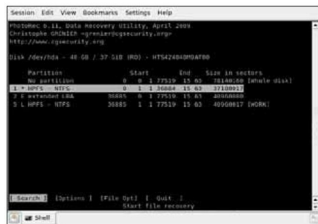


Figure 1: PhotoRec screen

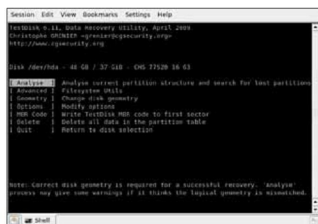


Figure 2: TestDisk menu